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# Co-creation as an approach to learning in architectural education

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# Manuscript:

# Co-creation as an approach to learning in architectural education - The hybrid practice of designing a temporary train station for the Roskilde Music Festival

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**Abstract**

The *purpose* of this paper is to explicate an educational approach to teaching architecture as a co-creation process, its case being the design and realization of a temporary train station for the Roskilde Festival 2025.

The *methodology* adopted in this paper is a case study of the built station deploying the degrowth evaluation matrix developed by Buch Hansen et al (2023), to assess sustainability parameters in both the teaching environment and the final result.

The *findings* point to lowered levels of stress and anxiety in the student group involved, while also highlighting pitfalls and fallbacks in relation to collective processes and individual learning outcomes.

The paper presents an *original* case, the case study of a building as a teaching object, while also presenting a hybrid co-creation between its authors from both economic studies (degrowth) and design studies (architecture).

Key words: degrowth, hybrid practice, co-creation, design for disassembly, architectural education, sustainability, case study

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## Introduction

Sustainability transformations require profound and deliberate changes across all domains of social life. Among other things, this includes fundamental changes to how and what we build, as building and construction makes up for large quantities of energy use and consumption of material (Sachs, 2015). Architectural education plays a critical role in preparing future architects to design and construct buildings that minimize environmental harm. In addition, through practices such as place-based learning and collaborative design, architectural education can help cultivate values, skills, and outlooks conducive to broader societal transformations (Engholm, 2023).

In the fall of 2024, a group of bachelor students of architecture from The Royal Danish Academy – School of Architecture, Design and Conservation undertook the task of designing a temporary train station for the Roskilde Festival, an annual music festival west of Copenhagen. The project was developed in close collaboration with the festival's coordination team and the Danish railway operator DSB. The station serves the western camping grounds of the festival and accommodates approximately 132,000 passengers, making it comparable in scale to busy transportation hubs nationwide. The design was guided by the Roskilde Festival's sustainability principles and realized through a co-creative process involving not only architecture students and joinery apprentices, but also volunteers, designers, user groups and professional builders. The final design emerged as a collective proposal developed by 25 students enrolled in the third-semester course of the bachelor programme "Complexity and Practice" at the Institute of Architecture, Urbanism & Landscape. Compared to traditional educational practice in design and architecture at the Royal Danish Academy, this departure stands out in two ways. First, in designing a project to be realized as a building with a context, programme and users. Secondly by students working as one collective producing one proposal, in the form of students as producers (McEvan, 2026), in contrast to common educational studio practice that is often carried out by individuals working on solitary design proposals.

In recent years, the concept of 'degrowth' has gained increasing traction within sustainability science and beyond. Advocates of degrowth argue that the use of resources in affluent societies must be radically reduced, requiring a departure from capitalist, growth-oriented economic systems (Latouche, 2009). UN IPCC reports on climate change (IPCC, 2023) and IAE UN reports (IEA, 2019) link the building sector in Europe alone with 36% of energy use and 39% of energy and carbon related emissions (Stilling, 2023). Degrowth theorists envision deeply transformed societies

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characterized by alternative modes of production and consumption, reconfigured work structures, gender equality, reduced exploitation, and stronger democratic institutions (Chertkovskaya et al., 2019; Schmelzer et al., 2022). Education is frequently cited as a critical site for enabling these transformations (Buch-Hansen et al., 2025; Liuzzo & Tsai, 2025).

This article applies the lens of deep transformations theory, (Buch-Hansen et al., 2024), to a real-world case of architectural education. Specifically, it addresses the following two questions:

> Q1: *How can a large group of students co-create a design proposal to realize a build project within architectural education?*
>
> Q2: *How can the outcomes and processes of such an educational approach be assessed in terms of collaborative and qualitative value by using the deep transformations theory of degrowth?*

To address Q1 and Q2, the Roskilde Festival Station teaching project is examined as a case study.

One written report and one evaluation capture key stages of the process. The written report is a field study on the existing festival station of 2024 produced during the annual festival prior to the teaching project and used to develop the 2025 project brief. The evaluation is submitted by students and conducted after examination but before the station's 2025 realization, following the academy´s study evaluation matrix (Brochmann, 2023b).

Combined with observations from faculty and project partners, the evaluation informs an analysis using the methodological framework of the deep transformations theory of degrowth (Buch-Hansen et all, 2023). In section 1.3 we outline methodological approaches to complex co-creation processes and motivate the methodological choice that we explicate in section 1.4.

**Scope**

The report and the evaluation contribute to a broader investigation across three consecutive station iterations (2025–2027). This paper addresses only the process of the first iteration (2025). The evaluation is limited by voluntary participation and potential response bias. It primarily served to calibrate teaching and group processes for the 2026 iteration. The 2025–2027 sequence should

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therefore be understood as an ongoing process of evidence development toward a pedagogy of co-creation. This paper constitutes the first of three studies progressively clarifying the case.

The present contribution is an interdisciplinary collaboration between architectural educators and social scientists specializing in degrowth theory and practice. The article proceeds as follows: In the background section, we introduce the Roskilde Festival and teaching practices at the Royal Danish Academy, our stance on architecture as a hybrid practice and our thoughts on the collaborative nature of architecture as well as introducing the *deep transformations theory of degrowth,* that is deployed in the analysis of the project. In section 2 we focus on Q1 by introducing the semester structure and pedagogical approach, in conjunction with qualitative data collected through questionnaires. In section 3 we analyse the case study through the means of the *deep transformations theory of degrowth.* In closing we discuss Q1 and Q2 in section 4, by relating our choice of methodology to our analysis and findings.

## 1. Context

### *1.1 Roskilde Festival*

Roskilde Festival (RF) has been a centerpiece on the European rock music festival map since its beginning in 1971. Inspired by the countercultural ethos of the 1970s and modeled after the 1969 Woodstock Festival in the United States, RF has grown from a small, local initiative into one of Northern Europe's largest music festivals. Today, it attracts over 130,000 visitors and volunteers annually, temporarily becoming Denmark's fourth largest city during its eight-days of tenure (Brorman et al, 2023). Roskilde Festival may be understood as a ritual of liberation and escapism taking place within a broader societal context marked by productivism, defined social norms, and restrictions. As such, the festival constitutes a temporary space for individual liberation and collective transgression (Wivel, 2009). However, RF can also be seen as a temporary lab or city, where it is possible to experiment with sustainable design practices, community building, and resource governance, seeking answers to urban design challenges (Brorman, 2023). Roskilde Festival Group, a collective of entities behind the non-profit event, has gradually adopted a range of social, economic, and environmental sustainability strategies.

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The student assignment for the train station design was anchored in the Roskilde Festival's sustainability guidelines, particularly those outlined in the "Impact" publication issued by the festival's governing body (Roskilde Festival Group, 2025). These include a set of "circular building principles" emphasizing lightweight construction, material limitation, and design for disassembly. The guidelines prioritize reuse, recycling, and upcycling of building materials wherever feasible, and call for responsible consumption where new materials are necessary. Furthermore, models of recirculation, lending, and leasing are explicitly favoured over one-off acquisitions. These principles framed the architectural brief provided to students by the festival and DSB and served as a central point of departure for the semester's design work. Since 1996 RF has accommodated a train stop located on a site adjacent to the railroad tracks intersecting the festival grounds. For this location, the brief from RF requested a design of a temporary train station with a minimum life cycle of three years 2025-2027, to be assembled on site within a week in the end of June, disassembled within a few days after the festival, and easily storable in the remaining 11 months of the year.

### *1.2 Traditional teaching practices at the Royal Danish Academy architecture education*

While the teaching format at the Royal Danish Academy has its origins in the Beaux-Arts tradition, the architecture education, as an independent institution of the academy has gradually evolved towards a liberal arts oriented teaching model. Each student is assigned a personal desk for the duration of the semester, and weekly one-to-one supervision takes place at the desks. The pedagogical approach is intentionally informal: rather than directing students through prescriptive instruction, teachers guide them along their individual trajectories, supporting the independent development of their projects. Over the past decade, group work has become increasingly common and in these cases, teachers meet with the entire group, often together with an additional instructor, to discuss the work collectively. Every two to four weeks, students and teachers participate in a shared critique session, during which students present their work and receive feedback from teachers and invited guests, and are encouraged to give comments to their peers. Throughout this process, students are supported to cultivate and rely on their own creative capacities.

Because their projects are rarely realized as built works, students engage less directly with real-world constraints such as tectonics, economics and feasibility. As a result, they often develop strong conceptual and creative skills but face a steep learning curve upon entering professional

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practice, where they must adapt to the corporate and technical realities of contemporary architectural practices. Although the curriculum includes compulsory internships and increasingly incorporates projects grounded in current societal and material conditions, the transition from academia to professional practice remains challenging for many graduates.

Historically, real-world projects have been discouraged at the academy due to concerns regarding authorship and credit. Given the academy's distinctive pedagogical model, where teachers and students co-develop work, any built outcome would formally require shared attribution, complicating the professional and legal frameworks surrounding architectural authorship. Furthermore, such projects disrupt the playing field among architectural practices by offering services for free.

While design–build programmes are well established within architectural education (Carpenter, 1997), exemplified by programmes such as SCE at Parsons School of Design, Design + Make at the Architectural Association School of Architecture, and Rural Studio of Auburn University, the project examined here has some distinguishing characteristics. The brief evolves during the course, involving clients with legislatively stringent and oftentimes conflicting demands that challenges purely architectural intuition and causes high complexity similar to real world projects. Additionally, concerning pedagogics, students were guided to operate within a radically democratic, flat organizational structure in which the project was developed collectively, ultimately producing a work with no identifiable singular authorship. Meanwhile, the project's outcome has significant public impact, serving approximately 132,000 passengers during the festival.

### *1.3 Architecture as a Pedagogy of Co-Creation*

In *Collaborations: Rethinking Architectural Design*, edited by Voigt, Graff and Ludwig (2023), the collaborative nature of architectural production is thematised under the editorial headline *How do we work together?* The editors cite the UNESCO Charter for Architectural Education, which stresses that architects should be trained as generalists to develop collaborative and relational skills (UNESCO–UIA, 2017). At the same time, they point to a discrepancy between this collaborative essence of the profession and established practices in institutions and research. As they note, many architecture schools continue to teach through a predominantly individual approach to design, and multidisciplinary teams remain relatively uncommon (Voigt, Graff & Ludwig, 2023). This

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illustrates a persistent gap between the collaborative essence of architectural design and its representation in discourse and educational contexts.

By contrast, design studies have developed an extensive methodological framework for collaboration and co-creation. From co-design and participatory design to transient design and regenerative design (Engholm, 2023), the design disciplines have articulated discourses and methods that architecture often appears to lack. It should be noted however that the term: ”co-creation”, has a specific meaning inside this discourse, namely the relation between consumer and producer in customizing a product.

Our understanding of co-creation builds on a distinction between hosting, partnership, and co-creation as progressively deeper forms of collaboration, following the framework proposed by the Royal Danish Academy (Holdt et al., 2021). This framework draws on the New Public Governance paradigm, Victor Pestoff’s work on co-production (Pestoff, 2018), and Jacob Torfing’s research on collaborative innovation (Torfing, 2016).

Within this model, *hosting* refers to ad hoc collaboration on clearly defined, one-off tasks. *Partnerships* involve a shared purpose, pooled resources, and a high degree of trust and shared risk among participants. *Co-creation* (Danish: *samskabelse*, sometimes described as *co-creation+*) adds a further dimension centred on experiential outcomes and open-ended processes (“no plan B”). It emphasizes the democratic involvement of target groups and participatory problem-solving aimed at generating new insights and learning among all participants. As Holdt et al. (2021, p. 8) argue, “co-creation carries the seeds of tomorrow's society and holds the potential to build new and better solutions to society's social, environmental and health challenges.”

Within this definition, delivering a product for a client, or even collaborating to provide a service for a client, does not in itself constitute co-creation. The case presented here, however, involved a collective of design students working with a real-life client and commissioner while simultaneously engaging user groups, volunteers, and universal design NGOs, and collaborating closely with builders and craftspeople throughout the process. The breadth of actors involved, and the participatory nature of the process align with the criteria outlined above and therefore qualify the project as an instance of co-creation.

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Looking back historically, architectural practice,at least in Denmark,has been strongly informed by traditions of co-creation, such as the co-operative movement and the folk high schools. Around 1900, the movement "Bedre Byggeskik" (Floris, 2005) exemplified this influence, while in the inter and post-war period practices such as "Fællestegnestuen" (Lokvig, 1990) and "Vandkunsten" (Weiss *et al*, 2021) made collaboration and co-creation a defining feature of their social and environmental progressive identity. Although the hierarchically organized office, with its claim to individual authorship, still dominates the architectural scene, horizontally organized collectives with alternative approaches to planning and thinking about architecture are increasingly emerging (Donat-Cattin 2022). In "Collective Processes" this trend is being generationalized with examples of practices like RaumLabor, Baukuh, A-A collective, as exponents of collective and collaborative ways of working and thinking:

> "Indeed, to speak of collaboration is in a certain sense speaking of architecture itself: one might argue that the creative process would not exist without interaction and therefore the necessity to acknowledge the common making of architecture and the collective intents behind it." (Donat-Cattin p. 11).

An explicit stance on collaboration can also be found in *Collaborations*, where a team of German educators (Glogar et al., 2023) under the heading *A Collaborative Approach for Urban Design Education in Light of the Climate Emergency* argue for collaborative approaches to urban design in response to the climate crisis. Referring to Steinitz, they suggest that collaboration is not only intrinsic to architecture, landscape and urban design but also indispensable as a cross-disciplinary matrix for innovation and change:

> "We must learn to understand that almost everything we do to change the landscape by design requires collaboration, whether with architects at the smaller scale, urban designers at the middle scale, geographers at the larger scale, and with engineers at all scales and layers, and bankers and government officials. […] If we are to work on society's most important needs, both specialist knowledge and skills of collaboration in designing are essential" (Steinitz, 2020, p. 3).

Of particular importance for this discussion of collaborative education in architecture is the taxonomy of collaborative intensity developed by Seve, Redondo and Sega (2022) as adapted by Glogar *et al* (2023) in their work. The framework proposes a taxonomy for case study analysis that

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allows for the classification, preservation and proposal of diverse and inclusive strategies, focusing on the tools employed in co-creative processes, the time they require, the spatial setting in which they occur, and the purposes they serve.

When applied to the design phase of the Roskilde Festival Station project, this taxonomy enables us to classify most of our educational activities as highly intense, since they frequently included external partners through workshops, client briefs, and building meetings:

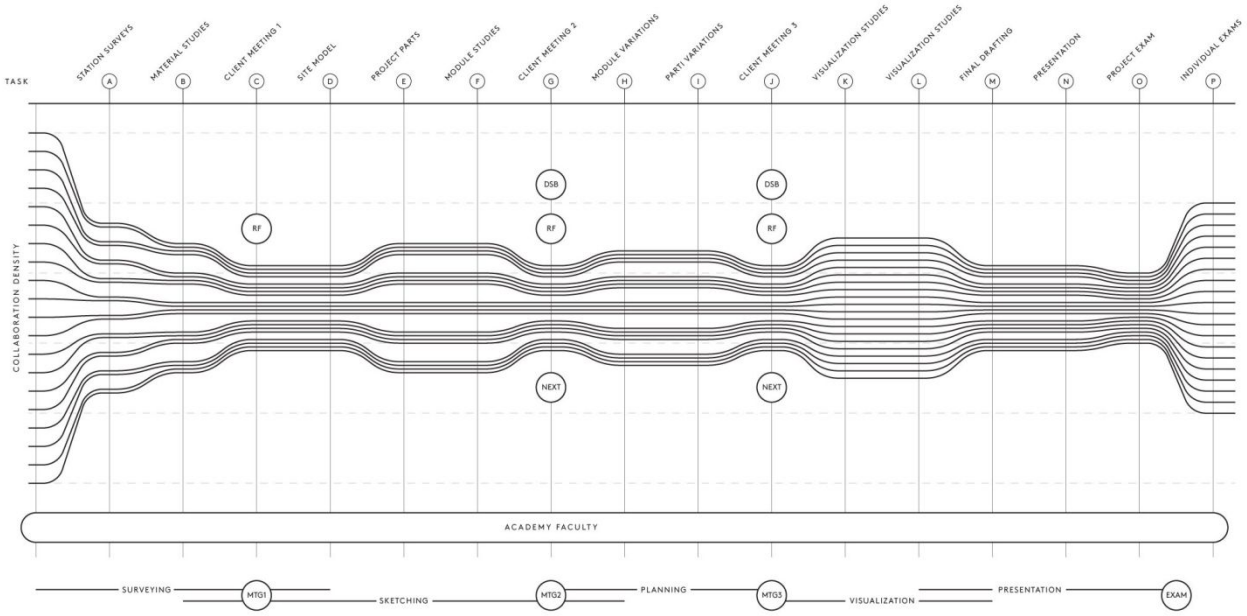



Fig 1 Process table showing modes of intensity of group work, freely adapted from Glogar et all (2023) (Case Study: Roskilde Festival Station 2025) Source: by the authors

However, intensity alone does not necessarily reflect the quality of collaboration. To establish a qualitative matrix, we turned beyond architecture to economic studies on degrowth. The evaluation matrix developed by Buch-Hansen and Nesterova (2023), originally intended to assess sustainability parameters in societal change, has thus been applied to examine whether our pedagogical approach not only increased collaborative intensity but also achieved qualitative goals—such as reducing stress while fostering sustainable, collective processes.

### 1.4. The deep transformations theory of degrowth

The Roskilde Festival station teaching project was not designed to be consistent with any theory or framework within degrowth. Yet in a chance encounter between members of the teaching team and social scientists working on degrowth, it became clear that undeniable compatibilities between the project and degrowth theorizing exist. Thus, in the next section we analyse the project through the lens of degrowth, specifically the deep transformations perspective. This perspective is especially pertinent to the current context as it allows for a holistic and nuanced analysis of the teaching case.

Deep transformations theory suggests that for societies to materialize in which humans are in harmony with themselves, fellow others, and nature, there needs to be less of some items characterizing contemporary societies (e.g., overproduction, overconsumption, competition,) and more of others (e.g., collaboration, empathy, justice). These items – derived from degrowth scholarship and related literatures – are referred to as, respectively, 'less items' and 'more items' (Buch-Hansen & Nesterova 2023). Change is understood to arise from the nurturing of the 'more items' that already exist in society in combination with transformation or even abolition of the 'less items'. Building on critical realist philosophy (Bhaskar 1993), the perspective argues that degrowth transformations – like any social phenomenon – need to be considered in relation to four planes of social being.

The planes are material transactions with nature, social interactions between persons, social structure, and peoples' inner being. 'Less items' and 'more items' are identified on each plane. These items are dialectically related, meaning that for each time there needs to be less of one thing, there needs to be a corresponding more of another (Nesterova et al. 2025). Entailing a process of 'absenting absences' (Bhaskar 1993), then, degrowth transformations entail replacing that which needs to be reduced ('less') with something different ('more').

Some examples, adapted specifically to the subject matter of design education, are provided below (Table 1).

**Table 1** Degrowth and education: selected 'less and more' items

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## 2. Case study

### *2.1 Semester framework*

The structure of the project was organized so that students of architecture collectively developed the architectural and artistic vision and produced the equivalent of a design handover to the client during a semester running from October 2024 to January 2025. In parallel, apprentices of joinery worked with mock-ups of the proposed design and eventually built the full structure at their campus, during spring 2025, before transporting all components to the Roskilde Festival site. The structure was then erected and positioned in the landscape, after which volunteers and architecture students joined to complete and finish it in time for the festival's opening ultimo June 2025.

Between the architectural students' handover in January and the project's completion in June, representatives from the festival assumed responsibility for obtaining municipal approvals, managing the budget, and making final adjustments to the design and siting in response to the ongoing development of the surrounding festival grounds. While the festival was engaged in these pursuits, the joinery apprentices, planned equipped and executed a production line capable of producing a large number of finalized A-frame modules from the initial mock-up and detailed drawings. In this process alterations of the base module concerning statics, buildability, workflow was implemented by them.

The teaching team's primary task was to develop a course structure that aligned the sustainability and architectural ambitions of the station project with the pedagogical aims of the semester itself. A key concern was to avoid creating a semester plan that demanded unrealistic investments of time and energy from students and teachers in the realization of a built structure.

The overarching theme of the third semester was materiality according to the curriculum of the BA program, with the course specifically titled Ny Roskilde Festival Station. The duration of the semester was 16 weeks (15 ECTS) and concluded with an oral examination and a pass/fail assessment. The 2024 semester included a cohort of 25 students and was led by three teachers with shared responsibility.

The project brief was developed in collaboration with representatives from Roskilde Festival and the Danish national railway operator (DSB), ensuring a real-world anchor and cross-institutional engagement. The background of the brief was a comprehensive report conducted by 10 students at

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the 2024 Roskilde Festival based on qualitative observations including interviews, site studies and general observations of the then existing station. This was carried out under the supervision of teachers and Roskilde Festival volunteers, and supplemented with information provided by the external partner DSB and comprised thematic chapters on zones and functions, structure, organization and users.

To structure the semester, the teaching team organized the course into three phases:

1. Contextual Study Phase (Weeks 1–4)
2. Sketching Phase (Weeks 5–10)
3. Project Planning Phase (Weeks 11–16)

A key pedagogical objective was to structure the process as a collaborative endeavour rather than a competitive exercise. To support this, students were divided into groups and asked to explore all assigned themes in parallel. This ensured that each group had a stake in the full spectrum of ideas, and that no single team "owned" a particular concept, thereby facilitating shared authorship and co-creation.

### *2.1 Client meetings*

The semester structure included three planned interactions with external stakeholders, referred to as "client meetings". At each meeting, students presented their ongoing design work and discussed future directions in dialogue with the external partners in a format between a normal review situation and a conversation with a client.

*Client Meeting 1:*

Objective: To present the client with a thematic approach to the structure and material palette of the project.

Themes: Wooden structure, Mobile structure, Ready-made structure, Tent structure, Landscape.

Organization: Five groups of five students. Each group produces one module of each of the five themes.

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Decisions: Together with the client it was decided to proceed with wooden structure and tent structure.

Key findings: Openness from external stakeholders towards artistic and abstract material. Relatively easy to establish common terminology in order to talk about architecture and problem solving in an inclusive way.

*Client Meeting 2:*

Objective: To present the client with a first concept of the organization of programs within a specific constructive method (tent or wood).

Themes: The A-frame module, the triangular module, the square module, the grid module, the organic module.

Organization: Five groups of five students, of which two groups split up and made alternate proposals.

Decisions: Together with the client it was decided to proceed with the principle of the A-frame module and the overall concept of a volume bending alongside the railroad tracks.

Key findings: Rapid evolution in student confidence towards speaking openly to external partners including in subjects where architectural ambition collide with real world functionality.

*Client Meeting 3:*

Objective: To present the client with a selection of proposals iterated from the module and landscape concept decided in meeting 2 and decision making for the final proposal.

Themes: Iteration of 1:20 study of the A-frame module from meeting 2.

Organization: Five groups of five students of which two split up and made proposals for furniture, landmark, and spatial configuration of the camping grounds.

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Decisions: Together with the client it was decided to follow a proposal that used the overall curved landscape concept, while incorporating aspects of the other proposals regarding lighting, landmark, spatial lay-out, meditation or meeting spaces. Final decision together with building partners and joinery apprentices on design proposals for A-frame module.

Key findings: A clear path towards the final phase of the project was established making the project planning phase feasibly with only minor adjustment to the overall proposed concept.



**Plate 1** Images from client meeting. From left to right: Table of proposal thematic structures from meeting 1. Presentation of conceptual volume studies from meeting 2. Presentation of proposed overall concept and A frame module from meeting 3. (Images credit: The Royal Danish Academy)

### *2.2 From concept to refinement: workshop and iteration*

Following the decision made in Client Meeting 3, the students participated in an intensive one-day workshop aimed at consolidating the agreed design direction while integrating key aspects from parallel proposals. The goal was to develop a joint 3D model that could accommodate insights and contributions across all student groups.

This collective 3D model served as the foundation for a two-week visualization workshop. During this period, joinery apprentices in parallel produced a 1:1 mock-up of the A-frame module, offering hands-on feedback and further informing the architectural development. Throughout this iterative phase, the teaching team introduced a series of minor design modifications. These included reducing the number of uniquely sized modules and introducing consistent radius dimensions in the plan layout. All changes were openly discussed in plenary sessions with the students to maintain transparency and collective authorship.

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### *2.3 Final Deliverables and Exhibition*

In the final stage of the semester, students were once again divided into groups, each responsible for producing specific components of the agreed deliverables for the client. These included:

- Site plan in 1:500
- Plan and sections in 1:100
- Construction details in 1:50, 1:20, 1:5
- Visualizations
- An updated digital 3D model
- Physical model 1:50

Two students were appointed curators, overseeing the layout and content of an exhibition that served both as the setting for the first day of the final examination and as the formal handover presentation to the client. They developed a display scheme, curated the spatial arrangement of presentation boards and models, and coordinated a four-hour joint presentation among the student cohort, rehearsed in advance. Each student was responsible for a 10-minute time slot focusing on a specific theme or deliverable.

A 1:1 mock-up, based on the revised module concept, was constructed by the joinery apprentices outside the examination hall. This full-scale element served as the culminating expression of the design process and a symbolic gesture of handover to the client.

On the second day of the examination, students were assessed individually. Each student presented a personal reflection, tracing their contribution and learning trajectory within the larger collective process. These reflections were documented in a "logbook" format, functioning as both an academic and introspective conclusion to the course.







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Plate 2 Images from examination + mock-up. From left to right: Day 01 of final examination exhibition format. Students presenting the 1:50 physical model during examination. Mock-up of the A frame module in the garden of the Royal Academy (Images credit: The Royal Danish Academy)

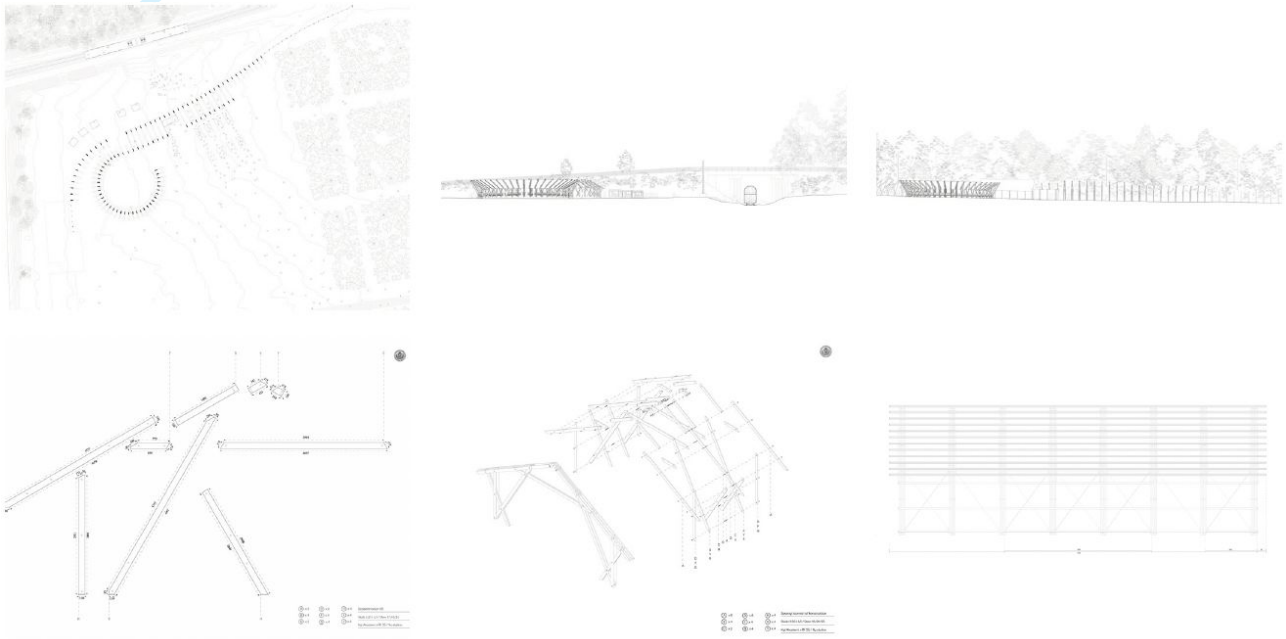


**Fig 2** Selection of delivered drawing material from the handover to the Roskilde Festival including plan, section, details and construction drawings (Source: The Royal Danish Academy)

### *2.4 Project description of the delivered proposal*

The project was designed as a "design for disassembly" modular scheme, allowing both flexibility in adaptation towards the landscape as well as reuse and storage of the temporary structure in between festival periods. Gently embedded in the surrounding landscape, the structure takes shape from the curvature of the existing railway tracks. This overall motif unfolds into three distinct typologies to assure recognition (Lechner, 2021):

- The House: A translucent archetype of shelter and comfort, offering protection from the elements. At night, it illuminates and serves as a natural wayfinding element, while also evoking a sense of familiarity and dwelling.
- The Portal: A welcoming gateway that marks the transition into the festival. Acting as a threshold, it symbolizes the passage from everyday life into the unique world of the festival experience.

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- The Circle: A central gathering space formed by a circle of benches, inviting rest and informal activity. This public area pays tribute to the ancient traditions of communal assembly in circular forms known in various scales from gathering around a fireplace to that of the Colosseum.



**Plate 3** Selection of model images of the 1:50 physical model (Images credit: The Royal Danish Academy)



**Plate 4** Selection of images from the finalized project in use. From left to right: The house and the circle at the time of arrival of the first train to the festival. Guest arriving from the portal into the festival ground on the outside of the communal space. A view from the open structure acting as waiting room until next arrival (Images credit: The Royal Danish Academy)

### *2.5 Reflections on Process and Pedagogical Values*

Reflections on the design process created specifically for this occasion have been ongoing. It is not customary at the Academy to realize student projects at full scale or to simulate office-like workflows. From the outset, the ambition was to create a project that would result in built architecture, without depleting the physical and mental resources of the students involved. This ambition faced two key challenges: first, the students were only in their third semester and thus had limited study-experience and knowledge; second, our internal evaluations of student work–life

balance at the Academy consistently reveal high levels of stress, anxiety, and overwork (Brockman, 2023b).

Several key observations emerged during the project. Students expressed frustration during group work aimed at building shared references and knowledge banks but also demonstrated the ability to produce convincing design proposals within short timeframes based on this material. Their presentations during "client meetings" were precise, focused, and to the point. Importantly, these meetings were consistently perceived as safer and more transparent than traditional in-house reviews.

Discussions in the meetings were typically grounded in objective criteria such as budget, constructability, disassembly, or storage. The meetings maintained a constructive and appreciative tone free from normative judgments or the tacit hierarchies that can characterize architectural education. As a result, students became comfortable presenting and debating ideas with external partners, which frequently led to collective decisions and a shared sense of ownership.

Stress mitigation was notably less in the final delivery stage of the project, where preparation, printing, and presentation normally peak. Students who were responsible for demanding drawings or spatial concepts were not responsible for layout or print production. Presentation duties were shared and rehearsed. No one stood alone at the podium, and no one arrived sleep-deprived or unprepared after an "all-nighter". It made a tangible difference: instead of preparing 25 individual submissions, only one single collective project was printed. This not only reduced the workload significantly, but it also allowed the Academy to cover production costs. The handover of the project thus became celebratory rather than exhausting.

### *2.6 Evaluation from the student body*

After the final examination students were encouraged to evaluate the course. A free text questionnaire for the students – to which 15 out of 25 students responded – was designed in compliance with academy rules and practices of anonymous evaluation and feedback (Brockman, 2023):

1. How has your learning been affected by having external critics and a client on the project?

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2. How have you experienced the shift in focus from an individual project to a collective result?

3. How have you experienced the way decisions in the process were made in dialogue between students and teachers?

4. Has the course allowed space for your individual academic/professional development?

5. Has your stress level and general well-being been above or below your experience from previous semesters?

6. How would you rate your overall satisfaction with the semester? *(1 = least satisfied, 5 = most satisfied)*

The number of respondents (60%) is considered valid, since the lower bar for respondents in the Royal Academy evaluations is 40% (Brockman, 2023a). As with other types of evaluation at the Academy, the least/most satisfied question hold one key of understanding, while the free text questions give a deeper qualitative insight into the student experience. An overall satisfaction score of 4 (3,88) on a 1-5 point scale corresponds to the threshold of interference and adjustment (below 80%) that standard evaluations instigate (Brockman, 2023a).

Of apparent interest are two key observations: 11 out 14 experienced that their level of stress was lower than usual, while 8 of these 11 respondents report a significant lowered stress and better thriving:

> My stress level has been significantly lower and my well-being much better. It has also made a huge difference in terms of stress and anxiety to know that we have been working together on the tasks. It has been thought-provoking for me to experience this huge difference in my well-being. I have the impression that I am not the only one in the class who has had this experience. I would like to encourage general reflection on how we can improve the learning environment on the "regular" semester projects in relation to this.

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For others the groupwork was demanding, partly because of planned circulation in groups, partly because of groups with a lower level of co-operation that lead to frustration. One respondent describes how this has led to a distance between individual and common project:

> This semester I think the shift has been drastic. Being thrown around in groups has made it difficult for me to find a foothold and a common basis and understanding. It has largely left me distanced from the project and with a lack of sense of ownership and initiative.

Many reported that working with real life clients gave them substantial learning benefits, thus 10 out of 14 students experienced that their learning outcome was enhanced by working with external partners and real-life parameters and restraints:

> For me, it has been positive. It has been educational to have to deal with their outside perspective on our work. Furthermore, it has been motivating to have to collaborate with a client in a more realistic way compared to the normal semester projects. It has also provided a lot of learning about communication between the parties during such a process.

While the other part, 4 out of 14, was very critical of lack of "traditional" studio feedback and evaluation formats:

> It has affected my learning to the extent that the usual *critique format* has been absent. A format that, in particular, serves as a learning space where I experience gaining deeper insight into my project and the process I am going through.

A small number felt that their individual growth or development as artists suffered. Delving into the question of collective versus individual learning and creative process, this point becomes more outspoken. Asked if the semester made space for individual learning one respondent answered the following:

> To a small extent, I would say**.** I feel that we've compromised individual learning a lot in favour of the collective result which, admittedly, can also be really good. However, I believe we weren't properly prepared for this task, precisely because we're only in our 3rd semester and that studying should first and foremost be

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about learning. Both technically and theoretically, but for me, it's also about learning to dare to dream and find one's own path in a field that's still completely new.

In overall terms, most (>80%) was satisfied, felt less stress and that their learning was enhanced by the process while others (<20%) felt that their individual learning suffered or that their stress Level increased due to shifts in group structure or dysfunctional groups.

## 3. Analysis

In what follows, we relate the Roskilde Station teaching case to the theoretical framework of deep transformation, going through the four planes of social being in turn (see Table 2 for a summary).

The analysis is based on a voluntary self-evaluation format, including five questions specific to the project and one overall scoring parameter. This evaluation was initiated and formulated by the semester coordinators, as the semester was not formally evaluated through the Royal Danish Academy's established evaluation system. The evaluation is used as indicative material and does not in itself provide sufficient evidence to be considered conclusive results. However, as the project will be repeated three times within the duration of the collaboration agreement between the partners, it is in the interest of both the authors and the project that the accumulated data is used to inform and improve subsequent iterations.

### *3.1 Material transactions with nature*

The course sought to empower students with the knowledge and practical tools needed to create sustainable, climate-responsive environments by focusing on using renewable and reusable materials. Concretely, the designed station structure came to be composed of layered 1x5 inch "off-the-shelf" timber boards, bolted together in a system that allows damaged parts to be easily replaced. Guided by the principles of Design for Disassembly, the station is built to be taken apart and reassembled, supporting a lifecycle across at least three consecutive Roskilde Festivals (2025–2027). The timber components – treated with natural linseed oil to extend durability – were designed so that they can be stored flat-packed in containers to protect them from weather and

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reduce environmental footprint. Designed without traditional concrete foundations, the station uses ground screws, to minimize impact on the landscape. This temporary, adaptable structure not only models circular design principles but also contributes to a broader material library, with components planned for reuse beyond 2027 in other festival constructions.



**Plate 5** Selection of images Next campus. From left to right: Students building a mock-up of the A-frame module. A large-scale stencil in order to mass produce the modules. (Images credit: The Royal Danish Academy)

Through the project, students gained direct experience in working with sustainable materials and designing with social and environmental responsibility in mind. These are key competencies for shaping inclusive, low-impact spaces for the future. However, while sustainability was a key design priority, other important considerations also influenced the decision-making process. For example, during the design process, the use of traditional wooden joinery was considered. However, both the engineer and the joinery apprentices advised against it, pointing out that it would result in significantly higher material consumption. Similarly, the idea of constructing the structure from reused materials was explored but ultimately dismissed. This decision was based on concerns regarding safety standards. New materials are certified to have certain load-bearing qualities – and moreover, using new materials ensured a more efficient workflow while guaranteeing that the structure can be safely reused across its intended lifecycle at the festival.

### *3.2 Social interactions between persons*

To facilitate the creation of a work of architecture that could be executed and accommodate festival guests and personnel, the course entailed alternative forms of delivery in that it went beyond traditional teaching to instead create a hands-on, collaborative learning environment. The project was structured around inclusive teamwork that brought together diverse participants, blurring the conventional lines between client, user, designer, and maker. Architecture students from the

academy met with joinery apprentices in building meetings to discuss the appropriateness of different solutions. Experimentation with co-creation was at the heart of the course. Through workshops involving users, universal design experts, and multiple stakeholders—such as Roskilde Festival volunteers, DSB representatives, SUMH and Copenhack Lab—the process centred on shared decision-making and inclusive design. The involvement of these varied actors not only enriched the project with multiple perspectives but also allowed students to understand and navigate the social and political dimensions of design. In this respect, many of the softer or dreamy aspects of the design process and evaluation, made way for a more hands on, pragmatic and realistic process, where issues of buildability, legislation, economy, material constraints and sustainability became important parameters. Since this way of organizing a semester was new to students , all activities were detailed into week plans, that were presented and discussed with the students at the beginning of each week with a brief and dialogue about tasks, strategies and objectives. The final examination, open to all collaborators, symbolized the project's collective ownership. To create public awareness of the project, it was decided to present it at GRASP, a two-day knowledge festival organized by the Roskilde Festival Group, that brings together art, talks, workshops, panel debates, and performances to explore sustainable change.

The complexity and pragmatic reality of the project, led to a tight time schedule, that had to prioritize collective discussion, client meetings and decision-making over individual teaching and feedback. In their evaluations of the course, some architecture students expressed that while they found these processes and meetings relevant, their individual learning and artistic development stagnated during this semester, while technical, knowledge and skill-based competencies were enhanced. Many students felt that the extensive groupwork and collective aim of producing one joint proposal relived them of individual performance pressure, while that in some cases pressure was shifted to individual workings and conflicts in “dysfunctional” groups. For some re-orientating in new groups during the semester produced a social strain in relating to new collaborators. This goes to show that co-creation and alternative forms of delivery can also have some negative outcomes and perhaps be disorientating to students who are used to “conventional” education. In other words, moving from “less items” to “more items” is not an unproblematic process.

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### *3.3 Social structure*

The course introduced a rethinking of architectural education by challenging social norms pertaining to individual performance, competition and hierarchy. The station was designed in a non-competitive way by 25 authors, eliminating traditional individual ownership to projects. The intention of this approach was to allow students to focus on learning rather than performance and rivalry, supporting a healthy balance between personal life and academic demands. Finally, a flat hierarchical structure reinforced the course's emphasis on shared learning and decision-making. Students worked in groups of five to prepare proposals for different aspects of the design brief, which were debated openly in plenary sessions. This co-created knowledge became part of a shared knowledge bank.

However, the project also illuminated the challenges of moving against the grain of a traditionally individualistic and competitive discipline. The field of architecture is known for its celebration of star designers, for a long-standing reliance on competition formats and for thriving on unpaid labour and aspirations for individual prestige. This state of affairs is mirrored in many educational settings in schools of design and architecture (Voigt, Graff & Ludwig, 2023). The Roskilde festival station course in many respects succeeded in challenging engrained structural features of architectural education; it also exhibited some of the challenges of transforming existing structures. Challenges for example assumed the form of conflicts within groups, individual dissatisfaction with learning outcomes, and concerns that personal contributions were being overshadowed or compromised by pragmatic decisions in building meetings. Additionally, the openness of the process occasionally led to tensions around transparency, particularly regarding meetings between clients and faculty that excluded students.

### *3.5 Inner being*

Recognizing the high stress levels documented in recent assessments of mental health among students at the Royal Danish Academy, the course was designed to tap into the human capacity for cooperation. In a recent survey almost 25% of the BA students state that they experience severe daily stress, while 47% of these mention lack of trust in own capabilities as a reason and 63% expectations to individual performance (Brockmann, 2023b). The idea to cope with this was simple: while one student alone might struggle under the pressure of a large-scale assignment, 25 students

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working together and supporting one another could achieve far more, and with less personal strain. Though the stress of realizing a physical structure remained, the course evaluation from the students shows that their overall stress-level was substantially reduced, compared to other semesters.

Despite this general reduction in overall stress levels, the collective approach also revealed important nuances. Not all students experienced the collaborative process in the same way. While some found relief in the shared responsibility, others experienced increased stress due to the demands of group work. This goes to show that different personalities respond differently to the collaborative setup: for some, working closely with peers is freeing, nurturing and motivating, fostering a sense of belonging and support; for others, it introduces new pressures, such as navigating group dynamics, negotiating differing visions, or worrying about individual development being overlooked. This variation in experience underscores that while collective learning models offer powerful tools for reducing stress and enhancing output, they need to also be sensitive to diverse needs and learning styles to ensure positive outcomes for all students.

To recapitulate, the analysis shows that, while the deep transformations framework was not used to design the teaching itself, many of the "more items" are present. At the same time, the analysis points to the challenges inherent in moving towards these items (Table 2).

**Table 2** Social planes analysis

## 4. Discussion & findings

With the exception of small-scale initiatives and projects in the niches and on the margins of consumer society, degrowth society does not (yet) exist, and many contemplations of this society unfold in academic spaces. This article focused on a real-life situation, designing a temporary train station for a music festival, characterized by an inherent tension. On one hand, the site of the project is an inherently unsustainable, even hedonistic event, a place of escapism from daily life within society. On the other hand, it is a space for experimentation, a living lab where the project can be executed in a manner that takes genuine sustainability into account. Seeking to design the project in the most sustainable way possible resulted in doing both the project itself and education differently. Sustainability and education are inherently interrelated: building a future characterized by working-

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together requires a different kind of learning. It requires collaboration, teamwork, mutual support, and multiple voices being heard. Both the project design and the pedagogical approach to it resonated deeply with the recent theory of degrowth, called the deep transformations theory of degrowth. In this article, we posed and answered two questions. The first one (*How can a large group of students co-create a design proposal in order to realize a build project within architectural education?*) targeted the pedagogical aspect of the project. The second question (*How can the outcomes and processes of such an educational approach be assessed in terms of collaborative and qualitative value by using the deep transformations theory of de-growth?)* investigated how we used the degrowth matrix to assess value in this educational format.

Our point of departure was that architecture conceived in an unsustainable way can never truly be said to meet a criterion for social sustainability. In this paper we have examined our own practice, both design wise and educational, through the lens of the degrowth matrix. This leads us to the following findings:

- Working as a collective significantly reduces workload and stress in the individual.
- Untapping the potential of the collective radically enhances the design ability and creative power.
- Stepping away from ideal cases, dealing with real-life situations, are imperfect and messy.
- Different personalities experience groupwork individually, with individual gains and loss.
- Students are highly aware of their individual artistic development while working in groups.
- Individual needs must be addressed in the process of collective output vs individual expression.

RQ1: We have found through our analysis that we were able to create a more sustaining learning environment, with a serious reduction in stress levels, anxiety and performance pressure. We also found that “untapping” the potential of the collective enhanced the creative power and design ability of the student body. Instead of 25 individual projects we arrived at one project with an exponential growth in complexity and artistic output level. The design for a transformation hub to accommodate 132.000 passengers is a task way beyond the scope of any 3rd semester student. The size in square

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meters, the complex logistics of arrival and departure, flow of passengers, additional programming and ticket purchase points towards the complexity of a more advanced study level. At the same time the artistic level and ambition of the project also resembled a much more advanced level. The final proposal having no single "author" or group of students attached to it, tapped into both the spirit of the festival itself, as well as connotations connected to early and much beloved examples of railway architecture from the beginning of the previous century as it also held a sleek reference to the hexagonal shape in the logo of the client and its brand colour red.

In this project the students held the role of designers themselves. Students designed the station, iterated a constructive module through trial and error and conceived of the overall treatment of landscape, while creating specific architectural motifs, like "the circle", "the portal" and "the house". The teachers served as facilitators, while clients delivered "critique" in evaluations shaped as so-called "client meetings". In section 2 we gave a detailed account of how this semester was organized, in order to conduct a qualitative analysis of process and outcome in section 3. Here two learning outcomes become evident: Cooperation and co-creation. The former could be appraised through the lens of the UNESCO charter as well as described as a competence according to the learning taxonomy of the Royal Academy, which divides learning into knowledge, skills and competencies (Royal Danish academy, 2023). As most architectural practice play out in cooperation with others, in design teams and cross disciplinary enterprises, we can only underscore the importance of focusing on creating a learning environment that supports the learning and practice of such a competence. However, inside a learning taxonomy like the one applied at the academy based on national and EU standards, we focus on knowledge, skill and competencies, while neglecting the importance of experience. Here we are pointing to the central experience of succeeding with others as co-creators, not only professionally and technically but also artistically. Perhaps the most valuable experience in this endeavour is that of having seen, felt, touched and moved your body inside a design you yourself have projected with others, while not having to suffer the stress, workload and responsibility of a professional.

RQ2: The final design and the realized building sought to live up to the circular building principles outlined by the festival adding to these a social sustainable criterium of feeling safe especially during nighttime. The station was designed for disassembly by students of architecture and built by apprentices joinery. This leads to a great emphasis on quality of individual components and craftsmanship. Perhaps our contribution to the seeming paradox of the festival as sustainable is

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namely this. A degrowth design is possible with the focus on deliberate design solutions and qualitative craftmanship. Roskilde's own ambition to become a sustainable festival might be seen in this light. Not to grow as a festival, but to grow the festival in a more sustainable way.

In our analysis we used the deep transformations theory of degrowth to analyse to which extend our design was applicable to the principles of the festival, as well as how our educational approach could be said to live up to a degrowth standard of education (Buch Hansen et all, 2024). As mentioned, we found reduced stress levels that should be read in relation to an increase in the complexity of deliverables, program, size and scope of the project. We found that the students had learning outcomes in pragmatic aspects of what a real-world project, with clients and partners, can entail in ways that are both messy and imperfect. We found that they had additional learning outcomes as regard to cooperative competencies as well as artistic, but we also found that some students suffered both in relation to social capital spent in shifting or dysfunctional working groups as well as in being frustrated over stagnation in their individual artistic development. The degrowth matrix showed us that these ailments are not to be taken lightly. Frustration with shifting groups could have been avoided, as well as the experience by students of “not being seen” or taken serious in their search for individual artistic expression. In this manner the theory adopted to analyse this case, seems both versatile and precise in that it can both hold inner and outer aspects of a diverse enterprise as realizing a building within an educational context. Applied to the temporary train station at Roskilde Festival, the degrowth matrix enabled a multi-perspective analysis that includes creators, users, and aesthetic evaluation. It allows the project to be assessed in terms of intentions and outcomes, revealing how collaborative design processes and resource-conscious construction engage with ecological and social concerns. In this case, the matrix highlights how co-creation and collective building practices reduced stress among students while enabling them to manage a complex infrastructure project. This approach also helps identify artistic, typological, and spatial qualities that emerge from these cooperative processes. Rather than reducing architecture to a question of degrowth alone, the framework clarifies how different values; social, ecological, and aesthetical, are distributed within the project. To this extent, the deep transformations theory of degrowth functions as a practical evaluation tool for understanding architectural practice within changing environmental and social conditions.

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**Plate 6** Image selection of the structure in use doing the festival period including a view from the camp grounds, a view from the circular communal space and the station at night. (Images credit: The Royal Danish Academy)

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**References**

Bhaskar, R. (1993) *Dialectic: The Pulse of Freedom*. London: Verso.

Brockman, H.N (2023a): Håndbog for kvailtetssikring, The Royal Danish Academy, accessed 11/06/2025: https://kglakademi.dk/sites/default/files/paragraph/field_downloads/Kvalitetsh%C3%A5ndbog_Enkeltsider.pdf

Brockman, H.N (2023b): Undervisningsmiljø vurdering (UVM), Royal Danish academy, accessed 11/06/2025: https://kglakademi.dk/sites/default/files/paragraph/field_downloads/UMV%202023%20-%20Danmarks%20Studieunders%C3%B8gelse%20-%20Akademi%20og%20uddannelser_0.pdf

Brorman, B.J., and Koefoed, E (2023): *Et fællesskab, En forening, Et Ritual, En tilstand, En By*. Roskilde Festivalgruppen. Roskilde-Festival.dk

Buch-Hansen, H., Koch, M., and Nesterova, I. (2024) *Deep transformations: A theory of degrowth*. Manchester University Press: Manchester.

Buch-Hansen, H. and Nesterova, I. (2023) Less and more: Conceptualising degrowth transformations. *Ecological Economics*, 205, 107731.

Buch-Hansen, H., Nesterova, I., & Nielsen, P. (2025). Envisioning an academia for degrowth transformations. *Globalizations*, 1-18.

Carpenter, W.J. (1997), *Learning by Building: Design and Construction in Architectural Education*, John Wiley & Sons.

Chertkovskaya, E., Paulsson, A., and Barca, S. (Eds.) (2019) *Towards a political economy of degrowth*. Rowman & Littlefield.

Carnelli, E., Marcolini, F., Marino, F., & Sousa Santos, R. (eds.) (2023). *Collaborations: Rethinking Architectural Design. Dimensions. Journal of Architectural Knowledge*, 3(5). Transcript Verlag. https://doi.org/10.14361/dak-2023-0501

Donat-Cattin, N. (2022) *Collective processes: Counterpractices in European architecture*. Basel, Switzerland: Birkhäuser.

Archnet-IJAR: International Journal of Architectural Research

Engholm, I. (2023) Design for the new world: From human design to planet design. Bristol: Intellect.

Floris, L. (2005) Bedre Byggeskik: Bevægelse og Bygninger. Frederiksberg, Holbæk: Thanning & Appel ; Holbæk Museum

Glogar, I., Boucsein, B., Kellhammer, M., Ruf, S. and Schmidt, M. (2023) A Collaborative Approach for Urban Design Education in Light of the Climate Emergency. Dimensions. Journal of Architectural Knowledge, Vol. 3 (Issue 5), pp. 143-166. https://doi.org/10.14361/dak-2023-0509

Holdt, M. B., Hygum, H., Gerber, H., Hunt, J. B., Boll, F. C., & Nielsen, S. T. (2021). Samskabelse eller samarbejde? Forskelle, fordele og fremgangsmåder. https://www.ingerfair.dk/wp-content/uploads/2021/04/ingerfair_Samskabeles-eller-samarbejde_online_2021.pdf

IEA (2019), Global Status Report for Buildings and Construction 2019, IEA, Paris https://www.iea.org/reports/global-status-report-for-buildings-and-construction-2019.

Impact - The Roskilde Festival Group's goals and results for sustainable development / 2024. Available at: https://cms.roskildefestivalgruppen.dk/media/3vjnmccp/rfg-aftryk-2024-dk-web.pdf (Accessed: 04 December 2025).

Intergovernmental Panel on Climate Change (IPCC) (ed.) (2023) 'Technical Summary', in Climate Change 2022 - Mitigation of Climate Change: Working Group III Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge: Cambridge University Press, pp. 51–148.

Latouche, S. (2009). Farewell to growth. Polity.

Lechner, A. (2021). Entwurf einer architektonischen Gebäudelehre (2. edition). Park Books.

Liuzzo, C., & Tsai, M. (2025). Crisis of Imagination: Integrating Degrowth Pedagogy for Sustainability Innovation in Business Schools. Futures, 103614.

Lokvig, J. (1990): Visioner og virkelighed ko-operative arkitekter og det sociale boligbyggeri. Achitectura nr. 12, 1990, pp. 95-125

McEwan C (2026), *The periphery as a critical project: towards an architectural pedagogy for the Anthropocene*. Archnet-IJAR: International Journal of Architectural Research, Vol. 20 No. 2 pp. 442–462.

Archnet-IJAR: International Journal of Architectural Research

Nesterova, I., Buch-Hansen, H., Suter, M., Rennstam, J., & Froese, T. (2025). Degrowth and business: Towards a holistic research agenda. Journal of Cleaner Production, 145614.

Pestoff, V. (2018). Co-production and public service management: Citizenship, governance and public services management. Routledge.

Royal danish academy, 2023: Studieordning for bacheloruddannelsen i arkitektur ved Det Kongelige Akademi – Arkitektur, Design, Konservering. https://kglakademi.dk/da/arkitektur/om-arkitektuddannelsen.

Sachs, J. (2015). The Age of Sustainable Development. Columbia University Press

Schmelzer, M., Vetter, A., & Vansintjan, A. (2022). The future is degrowth: A guide to a world beyond capitalism. Verso Books.

Seve, B., Redondo, E. and Sega, R. (2022) 'Urban co-creation taxonomy', Journal of Urban Design, 27(5), pp. 589–604. doi: 10.1080/13574809.2022.2053283.

Steinitz, C. (2020) 'On Landscape Architecture Education and professional practice and their future challenges', Land, 9(7), p. 228. doi:10.3390/land9070228.

Torfing, J. (2016). Collaborative innovation in the public sector. Georgetown University Press.

UNESCO–UIA (2017) Charter for Architectural Education. International Union of Architects.

Weiss, K.L. and Manelius, A.-M. (2022) Vandkunsten. Copenhagen: Arkitektens Forlag.

Wivel, U. (2009). Roskilde, Barok film, 93 min. https://ulrikwivel.com

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**Table 1:** Degrowth and education: selected 'less and more' items

| Planes | Less | More |
|---|---|---|
| *Material transactions with nature* | - Teaching students to see nature as a resource pool and only considering the economic costs of materials.<br>- Working with resource-intensive materials | - Empowering students with the knowledge and tools needed to create sustainable, climate-responsive, and socially inclusive environments.<br>- Working with sustainable materials |
| *Social interactions between persons* | - Lecture-based teaching<br>- Emphasis on individual performance | - Alternative forms of delivery<br>- Experimentation with co-creation/collaborative formats |
| *Social structure* | - Workaholism, Performance culture<br>- Hierarchies<br>- Emphasis on grades | - Work-life balance<br>- Process-oriented learning culture<br>- Flat structures<br>- Emphasis on learning |
| *Inner being* | - Anxieties and stress | - Peace of mind |

**Source:** Adapted from, and adding to, Buch-Hansen et al. 2025.

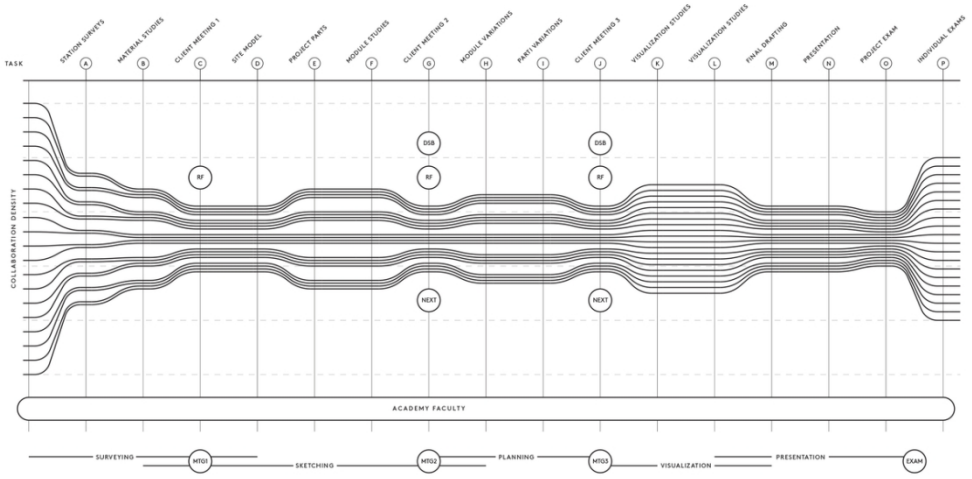
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**Table 2:** Social planes analysis

| Planes | Case |
|---|---|
| *Material transactions with nature* | The course focused on creating sustainable station design. For example: Timber structures treated with natural linseed oil. Emphasis on repairability, storability and reusability. "Design for Disassembly". Station was adapted to existing landscape. |
| *Social interactions between persons* | The course emphasized teamwork and co-creation. It aimed at erasing traditional borders between different actors in the building process (client, user, designer, craft persons, engineers, specialists). Involvement of wide range of stakeholders. Final examination open to collaborators. |
| *Social structure* | The course challenged social norms pertaining to individual performance, competition and hierarchy. Co-creation of solutions and of a shared knowledge bank, eliminating individual ownership of the project. |
| *Inner being* | The course tapped into the human potential for cooperation, aiming to reduce stress levels in the student cohort by lifting the burden off the shoulders of the individual student. |

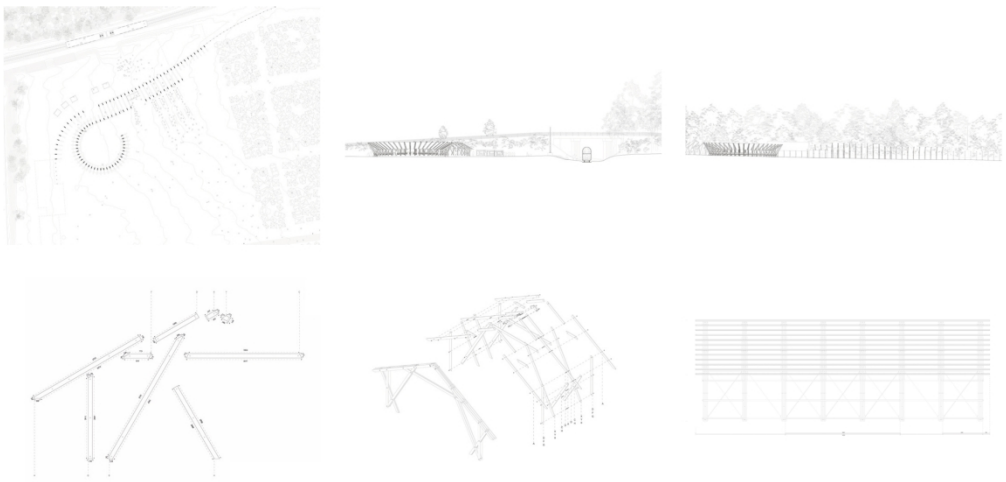
**Source:** Adapted from, and adding to, Buch-Hansen et al. 2025.

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Process table showing modes of intensity of group work, freely adapted from Glogar et all (2023) (Case Study: Roskilde Festival Station 2025) Source: by the authors

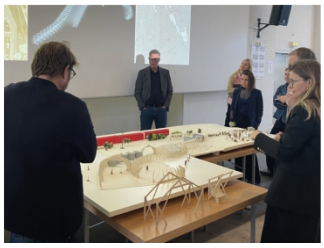
209x115mm (150 x 150 DPI)



Selection of delivered drawing material from the handover to the Roskilde Festival including plan, section, details and construction drawings (Source: The Royal Danish Academy)

487x236mm (144 x 144 DPI)





Images from client meeting. From left to right: Table of proposal thematic structures from meeting 1. Presentation of conceptual volume studies from meeting 2. Presentation of proposed overall concept and A frame module from meeting 3. (Images credit: The Royal Danish Academy)

564x135mm (144 x 144 DPI)



Images from examination + mock-up. From left to right: Day 01 of final examination exhibition format. Students presenting the 1:50 physical model during examination. Mock-up of the A frame module in the garden of the Royal Academy (Images credit: The Royal Danish Academy)

564x135mm (144 x 144 DPI)



Selection of model images of the 1:50 physical model (Images credit: The Royal Danish Academy)

565x135mm (144 x 144 DPI)



Selection of images from the finalized project in use. From left to right: The house and the circle at the time of arrival of the first train to the festival. Guest arriving from the portal into the festival ground on the outside of the communal space. A view from the open structure acting as waiting room until next arrival (Images credit: The Royal Danish Academy)

564x135mm (144 x 144 DPI)





Selection of images Next campus. From left to right: Students building a mock-up of the A-frame module. A large-scale stencil in order to mass produce the modules. (Images credit: The Royal Danish Academy)

564x134mm (144 x 144 DPI)




Image selection of the structure in use doing the festival period including a view from the camp grounds, a view from the circular communal space and the station at night. (Images credit: The Royal Danish Academy)

282x286mm (144 x 144 DPI)